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Combustion and emission characteristics in a gas turbine combustor at different pressure and swirl conditions

A. Datta *, [1], S.K. Som

Department of Mechanical Engineering, Indian Institute of Technology, Kharagpur-721 302, India

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Abstract

The practical challenge in research in the field of gas turbine combustion mainly centres around a clean emission, a low liner wall temperature, a uniform exit temperature distribution for turbomachinery applications along with a fuel economy of the combustion process. An attempt to meet up the challenge has been made in the present paper in developing a computational model based on stochastic separated flow analysis of a typical diffusion controlled spray combustion of liquid fuel in a gas turbine combustor to study the influences of combustor pressure and inlet swirl on the combustion and emission characteristics within the combustor. A standard k–ϵ model with wall function treatment for near wall region has been adopted for the solution of conservation equations in gas phase. The initial spray parameters are specified by a suitable PDF size distribution and a given spray cone angle. A radiation model for the gas phase, based on first order moment method, has been adopted in consideration of the gas phase as a grey absorbing–emitting medium. Formation of thermal NO_x, as a post combustion reaction process, is determined from Zeldovich mechanism.

It has been recognized that an increase in swirl number reduces the NO_x emission level and improves the pattern factor at all combustor pressures. However, though at lower pressure an increase in swirl number reduces combustion efficiency, the trend is exactly the reverse at higher pressure. With increase in pressure, pattern factor improves while the combustion efficiency falls and the level of NO_x emission increases. © 1999 Elsevier Science Ltd. All rights reserved.

Keywords: Gas turbine; Spray combustion; Emission

* Corresponding author. Tel.: +91-3222-55221; fax: +91-5222-55303.

[1] On leave from Power Plant Engineering, Department, Jadavpur University.

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Nomenclature

b	size parameter of Rosin–Rammler function
B	transfer number
c_p	specific heat
C_{f_s}	mean fuel vapour concentration at the droplet surface
C_j	mean concentration of jth species
d	droplet diameter
D_{eff}	effective mass diffusivity
$I_b(T)$	black body radiation intensity at temperature T
m	mass
$\dot{m}_{f_i}$	rate of fuel injection
M	molecular weight
Pr	Prandtl number
q_i^r	radiative heat flux in x_i direction
R	radius of the combustion chamber
Re_d	droplet Reynolds number
Re_{in}	inlet Reynolds number at swirler plane (ρ\rm $_{in}\cdot U_{z_{in}}\cdot D)/\mu$\rm $^{g}_{in}$)
$\dot{S}$	gas phase mass source term due to droplets
$\dot{S}_C$	gas phase species consveration equation source term
$\dot{S}_E$	gas phase energy source term
$\dot{S}_{M_i}$	gas phase momentum source term due to droplets
T	mean temperature
U_i	mean velocity in x_i direction
U_z	mean axial velocity
V_i	velocity in x_i direction
y	molar concentration of species
z	dimensionless axial location

Greek letters

α_{eff}	effective thermal diffusivity
γ	stoichiometric ratio
λ	stoichiometric coefficient
μ_t	eddy viscosity
μ_{eff}	effective viscosity
ρ	gas phase density

Superscript

d	droplet phase
g	gas phase

Subscript

f	fuel vapour
i	index of tensor notation

in	inlet plane
ex	exit plane
p	product species

1. Introduction

Extensive research works in the area of gas turbine combustion have gone on over the last two decades in order to improve the design of both stationary gas turbines for power generation and aero-propulsion gas turbines. The principal objectives of these research works are to improve the combustion process in order to achieve higher combustion efficiency with a uniform exit temperature and allowable liner wall temperature distribution and to reduce the combustion-generated polluting gases. Problems like acid rain and stratospheric ozone layer depletion increase the concern of today's world about the ill-effects of gaseous pollutants and have singled out NO_x as a potentially hazardous component in this regard. Control of NO_x emission from the combustion process thus becomes an important design criteria in modern gas turbine technology. A host of articles, both numerical and experimental in nature are available in literature on gas turbine combustion studies. The relevant numerical works [1–5] are based on the analysis of either axi-symmetric or asymmetric reacting flow in the combustor and predict mainly the flow structure and temperature profiles in different zones within the combustor along with NO_x concentration at combustor exit at different operating conditions of primary and secondary air flow and fuel–air distribution. The pertinent experimental works [6–10], on the other hand, focus on the influence of some important operating parameters like inlet swirl, air–fuel ratio, inlet pressure and temperature on flow and combustion characteristics in the combustor using both gaseous and liquid fuel. It has been recognized from the previous works that pollutant emission levels from the combustor are particularly sensitive to operating parameters like inlet air swirl and combustor pressure [6]. Most of the works reported in literature in relation to the effects of both inlet swirl [1, 3, 6] and pressure [5, 6] on the combustion process in a gas turbine type combustor were conducted with gaseous fuels. Numerical and experimental works available [2, 4] in relation to liquid fuel combustion in a gas turbine, on the other hand, do not provide an explicit information on the influences of inlet pressure and swirl on the combustion process. In liquid fuel combustion, combustor pressure will have a profound effect on the droplet vapourization rate by adjusting the partial pressure of fuel vapour at the droplet surface. The inlet swirl, on the other hand, will affect the trajectory of vapourizing droplets in determining the droplet penetration in the combustion chamber. Thus both these operating parameters influence the vapourization of liquid fuel and the subsequent mixing of fuel vapour with air and should have profound effects on the combustion process and the resultant temperature and concentration fields in the combustor fired with liquid fuel.

The present paper describes a numerical model of two-phase gas droplet flow in consideration of gas phase radiation to predict the influence of inlet swirl and combustor pressure on important combustion characteristics, namely combustion efficiency, temperature

distribution and NO_x formation caused by various complex processes associated with spray combustion in a gas turbine combustor.

2. Theoretical formulation

Evaporation and combustion of a continuously injected liquid fuel spray in a stream of air is solved in a can-type gas turbine combustor with divergent entry following a particle-source-in-cell (PSIC) modelling technique. The size and shape of the combustor (Fig. 1) is considered in conformity to the available literature. The air supply to the combustor is split among the swirler at entry and through two radial jets in the form of secondary and dilution air. The following assumptions are considered in the study:

(i) The problem is considered to be axi-symmetric. Hence the radial jets are taken to be uniform over the periphery.
(ii) Body forces and buoyancy forces are neglected.
(iii) Virtual mass force and Basset force on the liquid particles are not considered due to high density ratio between the two phases.
(iv) There is no nucleation, collision, break-up, coagulation and micro-explosion of droplets.
(v) Droplets do not take part in radiative energy exchange which is solved as a purely gas phase phenomena.
(vi) The fuel is considered to be *n*-hexane.

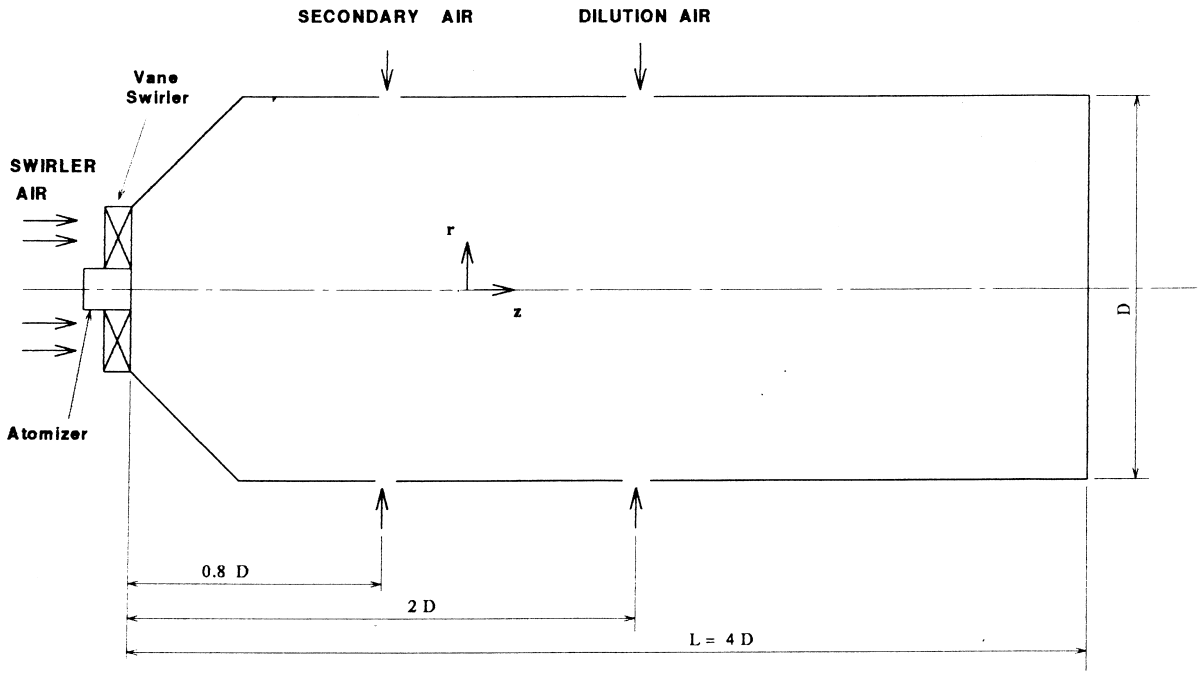

Fig. 1. Physical model of the problem.

2.1. Numerical model

The mathematical model comprises an Eulerian–Lagrangian formulation for the gas phase and droplet phase, respectively, in which the coupling between the two phases is taken care of through interactive source terms as described by Faeth [11].

2.1.1. Gas phase conservation equations

Following averaged gas phase conservation equations are solved:

Continuity:

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} + \rho\frac{\partial U_i}{\partial x_i} = \dot{S} \tag{1}$$

Momentum:

$$\frac{\partial}{\partial t}(\rho U_i) + \frac{\partial}{\partial x_j}(\rho U_i U_j) = -\frac{\partial p'}{\partial x_i} + \frac{\partial}{\partial x_j}\left\{\mu_{\mathrm{eff}}\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right)\right\} + \dot{S}_{\mathrm{M}} \tag{2}$$

where,

$$p' = p + \frac{2}{3}\rho k + \frac{2}{3}\mu_{\mathrm{eff}}\frac{\partial U_j}{\partial x_j} \tag{3}$$

and,

$$\mu_{\mathrm{eff}} = \mu + \mu_{\mathrm{t}} = \mu + c_\mu \rho \frac{k^2}{\epsilon} \tag{4}$$

Turbulent kinetic energy (k):

$$\rho\frac{\mathrm{D}k}{\mathrm{D}t} = \frac{\partial}{\partial x_i}\left(\frac{\mu_{\mathrm{t}}}{\sigma_k}\frac{\partial k}{\partial x_i}\right) + P - \rho\epsilon \tag{5}$$

where, turbulent kinetic energy production rate P is given by

$$P = \mu_{\mathrm{t}}\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right)\frac{\partial U_i}{\partial x_j} - \frac{2}{3}\frac{\partial U_i}{\partial x_i}\left(\rho k + \mu_{\mathrm{t}}\frac{\partial U_i}{\partial x_i}\right) \tag{6}$$

Turbulent kinetic energy dissipation rate (ϵ):

$$\rho\frac{\mathrm{D}\epsilon}{\mathrm{D}t} = \frac{\partial}{\partial x_i}\left(\frac{\mu_{\mathrm{t}}}{\sigma_\epsilon}\frac{\partial \epsilon}{\partial x_i}\right) + c_{1\epsilon}\frac{\epsilon}{k}P - \rho c_{2\epsilon}\frac{\epsilon^2}{k} \tag{7}$$

Typical constants for standard k–ϵ method are taken as:

$$c_\mu = 0.09, \quad c_{1\epsilon} = 1.44, \quad c_{2\epsilon} = 1.92, \quad \sigma_k = 1.0, \quad \sigma_c = 1.3$$

Individual species conservation

$$\frac{\partial}{\partial t}(\rho C_j) + \frac{\partial}{\partial x_i}(\rho U_i C_j) = \frac{\partial}{\partial x_i}\left(\rho D_{\text{eff}} \frac{\partial C_j}{\partial x_i}\right) + \dot{S}_{\text{C}} \tag{8}$$

The species equation is solved for fuel vapour, oxygen, carbon-dioxide and water vapour, while nitrogen concentration is obtained by difference.

Energy

$$\frac{\partial}{\partial t}(\rho_{\text{p}}^{\text{g}} T) + \frac{\partial}{\partial x_i}(\rho c_{\text{p}}^{\text{g}} U_i T) = \frac{\partial}{\partial x_i}\left(\rho c_{\text{g}}^{\text{g}} \alpha_{\text{eff}} \frac{\partial T}{\partial x_i}\right) - \frac{\partial q_i^{\text{r}}}{\partial x_i} + \dot{S}_{\text{E}} \tag{9}$$

The energy source term $\dot{S}_{\text{E}}$ has two components—one is the energy absorbed by the liquid drops during their heating-up period calculated from the interphase transport, and the other is the energy generated due to chemical reaction.

The reaction between fuel vapour and oxygen is assumed to be a single step, irreversible, global reaction following finite rate chemistry:

$$C_6H_{14} + 9.5O_2 \longrightarrow 6CO_2 + 7H_2O \tag{10}$$

The reaction rate is obtained from either of the following equations:

$$\dot{w}_k = K\rho^2 \frac{C_{\text{f}} C_{\text{O}_2}}{M_{\text{f}} M_{\text{O}_2}} \exp\left(-\frac{E}{RT}\right) \tag{11}$$

for kinetic controlled reaction

$$\dot{w}_{\text{d}} = A \frac{\rho}{M_{\text{f}}} \frac{\epsilon}{k} \min\left(C_{\text{f}}, \frac{C_{\text{o}_2}}{\gamma}, \frac{c C_{\text{p}}}{1+\gamma}\right) \tag{12}$$

for turbulent diffusion controlled reaction [12]. A and c in Eq. (12) are constants.

The energy source due to reaction will be,

$$\Delta H_{\text{R}} M_{\text{f}} [\min(\dot{w}_k, \dot{w}_{\text{d}})] \tag{13}$$

where ΔH_{R} is the enthalpy of reaction.

The radiative energy exchange is evaluated assuming the gas phase to be a grey, absorbing–emitting medium and solving the radiative transfer equation following the first-order moment method [13] which reduces the integro-differential equation of radiative exchange into a differential equation. The following equation is obtained for q_i^{r}:

$$\frac{\partial q_i^{\text{r}}}{\partial x_i} = \kappa[4\pi I_{\text{b}}(T) - G] \tag{14}$$

where the irradiation G is related to q_i^{r} as

$$q_i^{\mathrm{r}} = -\frac{1}{3\kappa}\frac{\partial G}{\partial x_i} \tag{15}$$

In the light of the assumption of a grey gas radiation model, the absorption coefficient κ is replaced by Planck's mean absorption coefficient κ_{P}. The local value of κ_{P} is computed as,

$$\kappa_{\mathrm{P}} = C_{\mathrm{H_2O}}\kappa_{\mathrm{P_{H_2O}}}(T) + C_{\mathrm{CO_2}}\kappa_{\mathrm{PCO_2}}(T) \tag{16}$$

The values of $\kappa_{\mathrm{P}}(T)$ as a function of temperature for CO_2 and H_2O are taken from Sparrow and Cess [14].

2.1.2. Generation of droplet phase information

The fuel spray injected continuously into the combustor is considered to be specified by an initial size distribution function along with an initial velocity and temperature same for all droplets. The velocity, mass and temperature history of all droplet classes along their trajectories are obtained from the respective conservation equations on a Lagrangian frame as follows:

Droplet velocity

$$m^{\mathrm{d}}\frac{\mathrm{d}V_i^{\mathrm{d}}}{\mathrm{d}t} = \frac{\pi}{8}C_{\mathrm{D}}^2|V_i^{\mathrm{g}} - V_i^{\mathrm{d}}|(V_i^{\mathrm{g}} - V_i^{\mathrm{d}}) \tag{17}$$

Drag coefficient C_{D} is computed following the standard drag law [15].

The effect of gas phase turbulence on the droplet motion is simulated using a stochastic approach. The instantaneous gas phase velocity (V_i^{g}) is obtained by computing the fluctuating velocity component from the turbulent kinetic energy in consideration of isotropic turbulence, and using a normally distributed random number ζ as

$$V_i^{\mathrm{g}} = U_i + \zeta\sqrt{\frac{2k}{3}} \tag{18}$$

Droplet mass

$$\frac{\mathrm{d}m^{\mathrm{d}}}{\mathrm{d}t} = -\rho^{\mathrm{m}}\beta\pi d^2(C_{\mathrm{f_s}} - C_{\mathrm{f}}) \tag{19}$$

where, ρ^{m} is the density of gas phase at the droplet surface.

Droplet temperature

$$m^{\mathrm{d}}c_p^{\mathrm{d}}\frac{\mathrm{d}T^{\mathrm{d}}}{\mathrm{d}t} = h\pi d^2(T - T^{\mathrm{d}}) + \frac{\mathrm{d}m^{\mathrm{d}}}{\mathrm{d}t}\Delta H_{\mathrm{v}} \tag{20}$$

where, ΔH_{v} is the enthalpy of vapourization of the liquid fuel at droplet temperature.

Mass transfer coefficient β and heat transfer coefficient h in Eqs. (19) and (20) are evaluated from the standard correlations

$$Sh(1+B) = 2 + 0.6Re_{\mathrm{d}}^{0.5}Sc^{0.33} \tag{21}$$

$$Nu(1+B) = 2 + 0.6Re_{\mathrm{d}}^{0.5}Pr^{0.33} \tag{22}$$

Eqs. (17), (19) and (20) are solved for V_i^{d}, m^{d} and T^{d} with appropriate initial conditions.

The initial drop size distribution of liquid fuel spray is assumed to follow a realistic Rosin–Rammler four parameter distribution function, given as,

$$G'(d_i) = \frac{\exp(-bd_i^{\mathrm{n}}) - \exp(-bd_{\max i}^{\mathrm{n}})}{\exp(-bd_{\min i}^{\mathrm{n}}) - \exp(-bd_{\max i}^{\mathrm{n}})} \tag{23}$$

where $G'(d_i)$ is the mass fraction of the spray having diameter above d_{in}. The dispersion parameter n is taken as 3 following Mugele and Evans [16].

2.1.3. NO formation

As NO constitutes the largest share of NO_x formed in combustion, its formation is calculated following the Zeldovich mechanism in a similar manner to that adopted by Ramos [1]. Since the initial step of this reaction mechanism involves a very high activation energy, it is assumed that the NO formation process is decoupled from the hydrocarbon combustion and occurs as a post-combustion reaction process. The NO formation is governed by the following equations:

$$\mathrm{O} + \mathrm{N_2} \rightleftharpoons \mathrm{NO} + \mathrm{N} \tag{24}$$

$$\mathrm{N} + \mathrm{O_2} \rightleftharpoons \mathrm{NO} + \mathrm{O} \tag{25}$$

A steady-state assumption made for the calculation of N-atoms gives

$$\frac{\mathrm{d}y_{\mathrm{v}}}{\mathrm{d}t} = 0 \tag{26}$$

The O-atom is assumed to be in equilibrium with the O_2 molecule as

$$\mathrm{M} + \mathrm{O_2} \rightleftharpoons \mathrm{M} + 2\mathrm{O} \tag{27}$$

where M is a third body.

The rate of formation of NO is calculated as

$$\frac{\mathrm{d}y_{\mathrm{NO}}}{\mathrm{d}t} = K_1 y_{\mathrm{N_2}} y_{\mathrm{O}} - K_2 y_{\mathrm{NO}} y_{\mathrm{N}} + K_3 y_{\mathrm{N}} y_{\mathrm{O_2}} - K_4 y_{\mathrm{NO}} y_{\mathrm{O}} \tag{28}$$

where the values of y_{N} and y_{O} are obtained from the assumptions given in Eqs. (26) and (27). The specific reaction rates K_1–K_4 for the above reactions are taken from Ramos [1].

The species conservation equation for NO is solved with its rate of formation as the source term to get NO concentration within the combustor as well as at the combustor exit.

3. Method of solution

3.1. Numerical scheme

The numerical code consisted of two different modules—one for gas phase computation and the other for calculation of the droplet phase.

The gas phase conservation equations were solved with appropriate boundary conditions following explicit finite difference technique. The conservation equations were first non-dimensionalised against reference quantities and then discretized following a staggered grid arrangement. The reference length was taken to be the combustor diameter (D), the reference velocity to be the axial velocity through the swirler ($U_{z_{in}}$), the reference temperature to be the ambient temperature ($T_a = 300$ K) and the reference time to be ($D/U_{z_{in}}$). The diffusion and convective terms were discretized by central and hybrid differencing schemes respectively. The coupling between pressure and velocity was accomplished using a MAC-based SOLA algorithm [17] . Energy and species conservation equations were solved in a coupled form with the continuity and momentum equations to account for the variation in density with temperature and concentration. The solution procedure in this module continued till a steady state convergence was attained.

The ordinary differential equations of the droplet phase were solved by a fourth order Runge–Kutta method to generate droplet information required for the gas phase source terms. The solution in this module continued until all the representative droplets of the different classes extinguished in the process of evaporation.

The two modules were invoked alternately to solve the respective equations until a final convergence was arrived at. This was considered to be achieved when the result of any module did not alter the result of the other module.

3.2. Numerical mesh

A numerical mesh of 85×30 grid nodes was used after several experiments, which showed that further refinement in either direction did not change the result (maximum difference in velocity and other scalar functions in the carrier phase) by more than 2%. The grid spacings in axial and radial directions were changed smoothly to minimize the deterioration of the formal accuracy of the finite difference scheme due to variable grid spacing and in such a way that a higher concentration of nodes occurs near the swirler and the radial holes.

3.3. Operating parameters and boundary conditions

The total air flow to the combustor was split among the swirler and the two radial jets according to a ratio of 5:7:8 following Cameron et al. [10]. An air–fuel ratio of 60 was used. Swirl number was calculated assuming a solid body type rotation generated by the helicoidal vanes in the swirler. The air entering the combustor was considered to be at

600 K. The fuel spray is considered to have a sauter mean drop diameter of 52 μm and a cone angle of 80°.

A zero axial gradient was prescribed at the outlet for all variables. Standard logarithmic law of wall was considered at the solid walls. For radiation calculation, the end planes were assumed to be radiatively adiabatic and Marshak boundary condition was applied at the solid combustor wall.

4. Results and discussion

4.1. Flow and temperature distribution in the combustor

Figs. 2a and b show the flow field within the combustor under reactive situation for two different inlet swirl numbers. At a higher swirl number of 0.76 (Fig. 2a) a distinct toroidal recirculation is observed at a central region in the primary zone of the combustor. The recirculatory flow is shifted slightly off-axis at the downstream near the location of the secondary jet injection. However, at a lower swirl number of 0.37 (Fig. 2b) the central

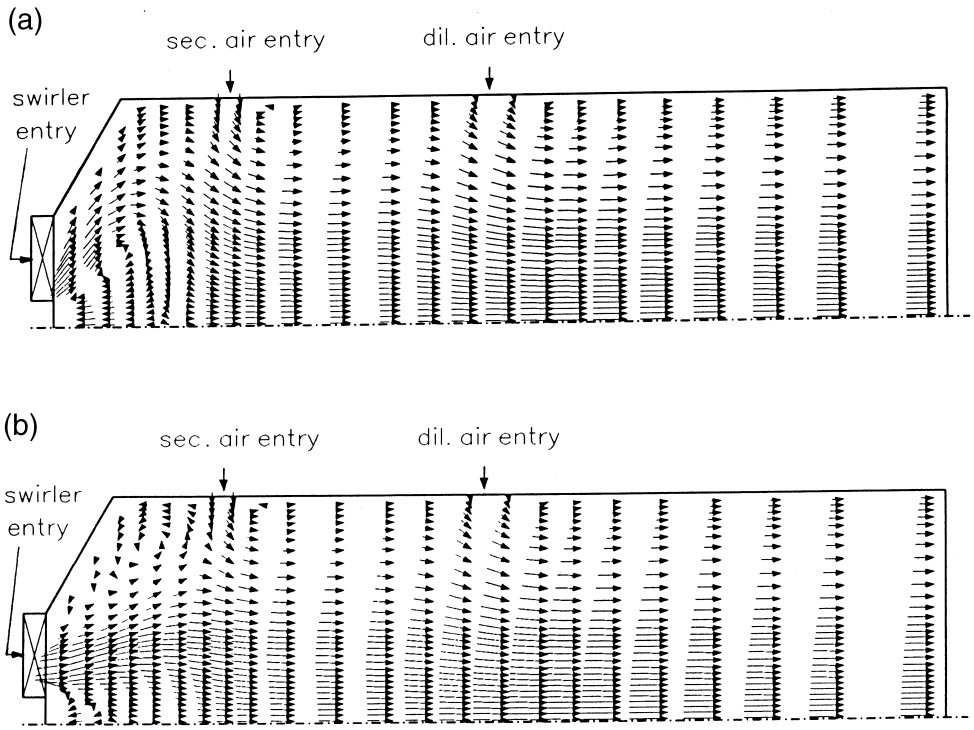

Fig. 2. Velocity vector plot within the combustion chamber, (a) $Re_{in} = 52{,}100$, swirl number = 0.76, combustor pressure = 100 kPa. (b) Re_{in} = 52,100, swirl number = 0.37, combustor pressure = 100 kPa.

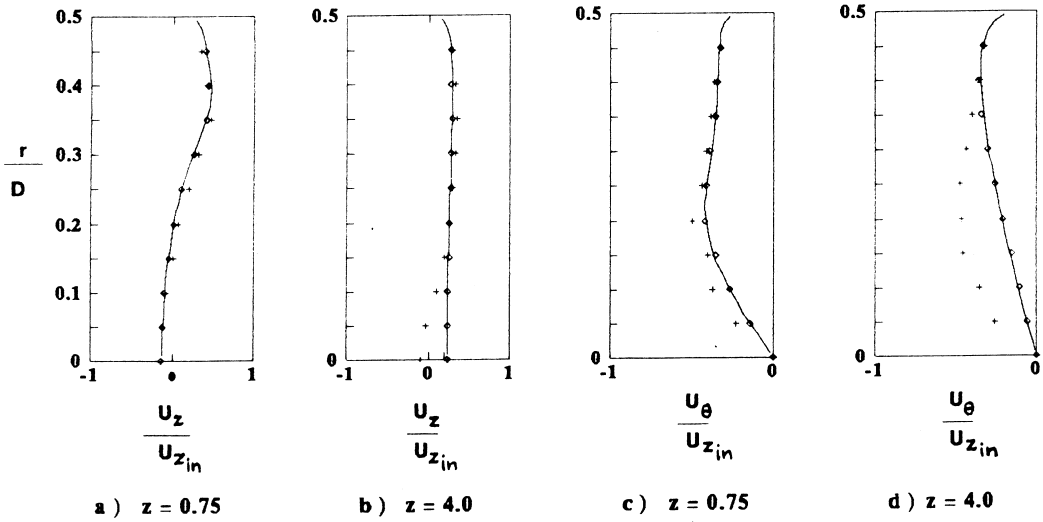

Fig. 3. Comparisons of predicted axial and tangential velocity distributions with Chang and Chen [18] and Dellenback [19]. (—) Present computation, (◇) Chang and Chen, (+) Dallenback.

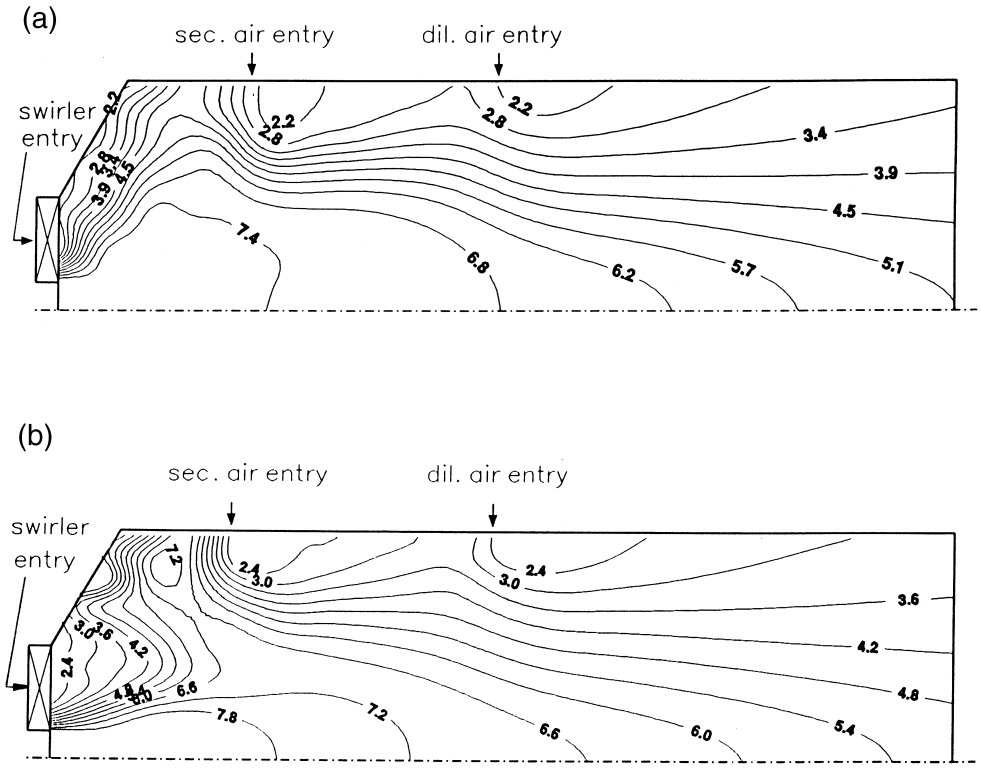

Fig. 4. Contours of dimensionless temperature (T/T_a) within the combustion chamber. (a) $Re_{in} = 52{,}100$, swirl number = 0.76, combustor pressure = 100 kPa. (b) $Re_{in} = 52{,}100$, swirl number = 0.37, combustor pressure = 100 kPa.

recirculatory zone is absent, while a recirculation is observed around the wall region in the primary zone which may be attributed to the expansion that occurs as the flow enters the chamber. The radial pressure gradient for higher swirl number overcomes this wall recirculation in the flow.

A comparison of the axial and tangential velocity distribution predicted by our model, in case of an axi-symmetric sudden-expansion swirling flow, has been made with the computational work of Chang and Chen [18] and the experimental results of Dellenback [19] under the similar situation in order to calibrate the present model for the elliptic flow field. Figs. 3a and c show a fair agreement of the predicted axial and tangential velocity profiles near the inlet region ($z = 0.75$) with the computations of Chang and Chen, and the empirical values of Dellenback. However, it is observed from Fig. 3b and d, that the present numerical results and the computations of Chang and Chen, being in agreement with each other, do not conform well with the experimental results of Dellenback for the tangential velocity distribution at far downstream from the inlet ($z = 4.0$) and mainly near the axis. This was explained by Chang and Chen in the light of under-prediction of the length of central toroidal recirculation zone by the standard k–ϵ model. Therefore, the theoretical model predicts a relatively lower strength of swirl at the downstream location.

Figs. 4a and b describe the dimensionless temperature distribution within the combustor under a steady reactive condition for the given operating parameters. The high temperature contours, depicting the flame, are located in the primary zone from where they spread downstream. For a higher swirl number (Fig. 4a) the flame, guided by the typical flow field,

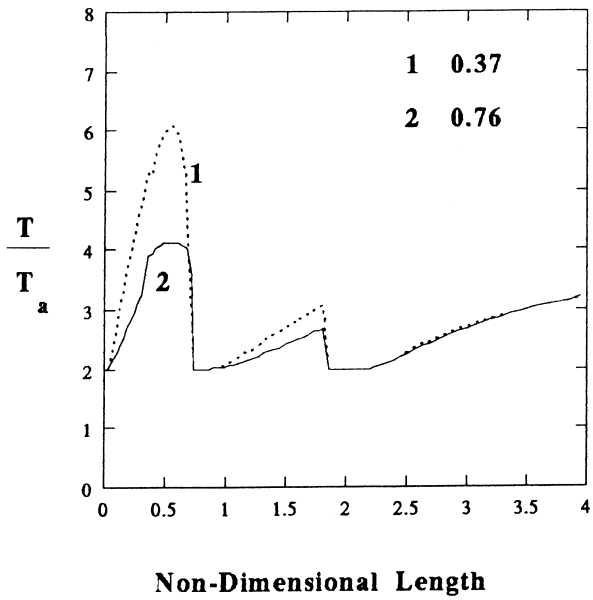

Fig. 5. Variation of liner wall temperature along the length of the combustor for different swirl numbers (combustor pressure = 100 kPa).

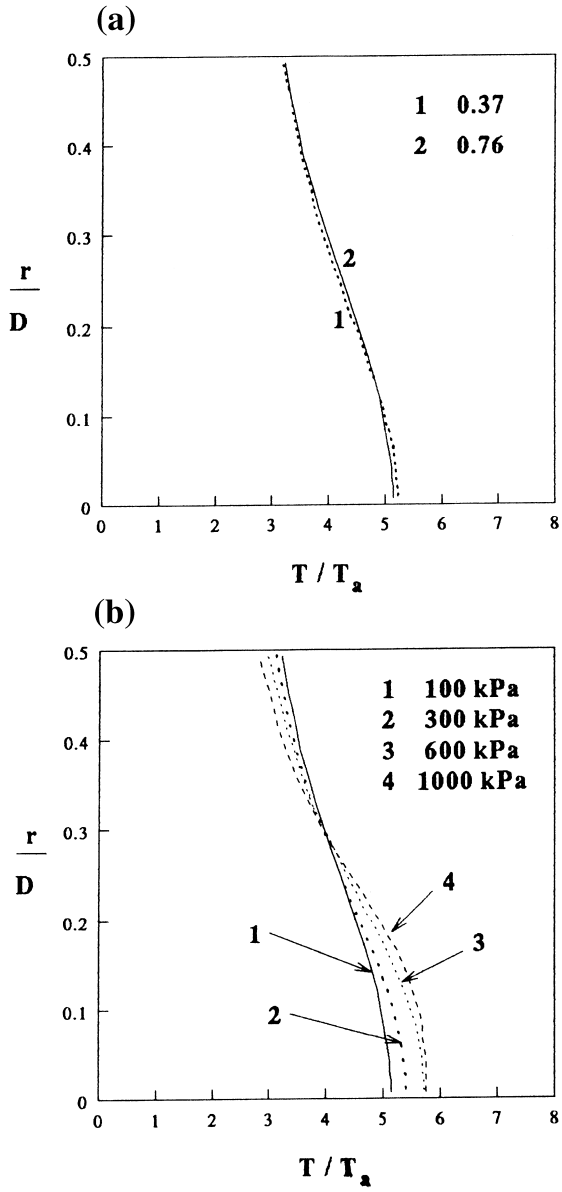

Fig. 6. (a) Exit temperature distribution for different swirl nmbers ($Re_{in} = 52{,}100$, combustor pressure = 100 kPa). (b) Exit temperature distribution for dfferent combustor pressure ($Re_{in} = 52{,}100$, swirl number = 0.76).

Table 1
Influence of inlet swirl number on combustion and emission parameters. Inputs: $Re_{in} = 52{,}100$, mean drop diameter = 52 μm, spray cone angle = 80°, air–fuel ratio = 60, inlet temperature = 600 K

Sample number	Inlet pressure (kPa)	Swirl number	Combustion efficiency (%)	Pattern factor	Exit NO concentration (bulk) (ppm)
1	100	0.37	97.59	0.710	72.37
2	100	0.76	95.34	0.656	19.16
3	600	0.37	87.24	1.062	250.8
4	600	0.76	91.32	0.888	203.7

spreads in the radial direction, but shows a lower temperature near the wall. On the other hand, for a lower swirl number, the flame, in the absence of the central recirculation zone, elongates in the axial direction and at the same time engulfs the region near the wall due to the wall recirculation in the primary zone. This causes a higher liner metal temperature for a lower inlet swirl. The temperature distribution along the combustor wall (Fig. 5) depicts the physical phenomena as well.

4.2. Exit temperature

The combustor exit temperature distribution and the corresponding pattern factor have a strong impact in a gas turbine plant. Fig. 6a and b show the effects of inlet swirl and combustor pressure on the exit temperature distribution, corresponding pattern factor values are listed in Table 1 and 2. It is observed under all situations that the exit temperature follows the same pattern with a maximum at the axis and the lowest value near the wall, which may not be true in the asymmetric combustion chambers with discrete blowing holes for actual gas turbines. However, the predicted bulk temperature from our model agrees well

Table 2
Influence of combustor pressure on combustion and emission parameters. (Inputs as Table 1)

Sample number	Inlet pressure (kPa)	Swirl number	Combustion efficiency (%)	Pattern factor	Exit NO concentration (bulk) (ppm)
1	100	0.76	95.34	0.656	19.16
2	300	0.76	93.56	0.813	125.2
3	600	0.76	91.32	0.888	203.7
4	1000	0.76	83.31	0.923	310.7

with the counterpart experimental results by Cameron et al. [10] under a similar situation. Fig. 6a shows that inlet swirl number has almost a negligible influence on the exit temperature distribution. Combustor pressure, on the other hand, shows a significant effect on the exit temperature distribution (Fig. 6b), with pattern factor values increasing with pressure. The non-uniformity in the exit temperature distribution with pressure may be caused by the fact that the increase in pressure hinders the liquid droplets from spreading across the combustor in the radial direction downstream. This will arrest the fuel vapour concentration mostly near the axis and will narrow down the high temperature zone in the central region attributing a non-uniformity in the exit temperature distribution.

4.3. Combustion efficiency

Combustion efficiency of a combustor is a measure of its ability to burn the fuel completely within it and is given by

$$\eta_c = 1 - \frac{2\pi \int_0^R \rho_{ex} \cdot U_{z_{ex}} \cdot C_{f_{ex}} \cdot r \cdot dr}{\dot{m}_{f_i}} \tag{29}$$

Thus the exit fuel vapour concentration becomes the index of combustion efficiency for a particular set of operating parameters and flow situation. Fig. 7a–c compares the exit distribution of fuel vapour under various operating parameters and Tables 1 and 2 give the corresponding η_c values. The most interesting feature observed in this context is that the qualitative trend of the influence of inlet swirl number of primary air flow on combustion efficiency becomes exactly the opposite when the combustor pressure is changed from a lower to a higher value. It is found that at a lower pressure of 100 kPa, an increase in inlet swirl number reduces the combustion efficiency, while the trend is reversed when the pressure is increased to 600 kPa. A possible physical explanation of this contrasting phenomenon may be put forward as follows: the purpose of imparting swirl to incoming primary air is to cause an on-axis recirculatory flow near the upstream region of the primary zone in enhancing the rates of droplet vapourization and mixing of fuel vapour in air along with the subsequent stabilization of flame, but on the other hand the recirculatory flow field in the inlet region of the combustor reduces the penetration of droplets to further downstream in the primary zone for a more effective droplet vapourization and burning. The relative role of these two counter-weighing effects depend on the range of operating parameters, mainly the inlet pressure and mean drop diameter. At lower pressure the adverse effect of reduced droplet penetration due to an increase in inlet swirl of primary air plays the dominant role in reducing the combustion efficiency. However, when the pressure in the combustion chamber is high, the droplet penetration is as such low. Hence, the favourable effect of enhanced droplet vapourization and mixing of fuel vapour in air on the burning rate due to an increase in inlet air swirl plays the prominent role in increasing the combustion efficiency.

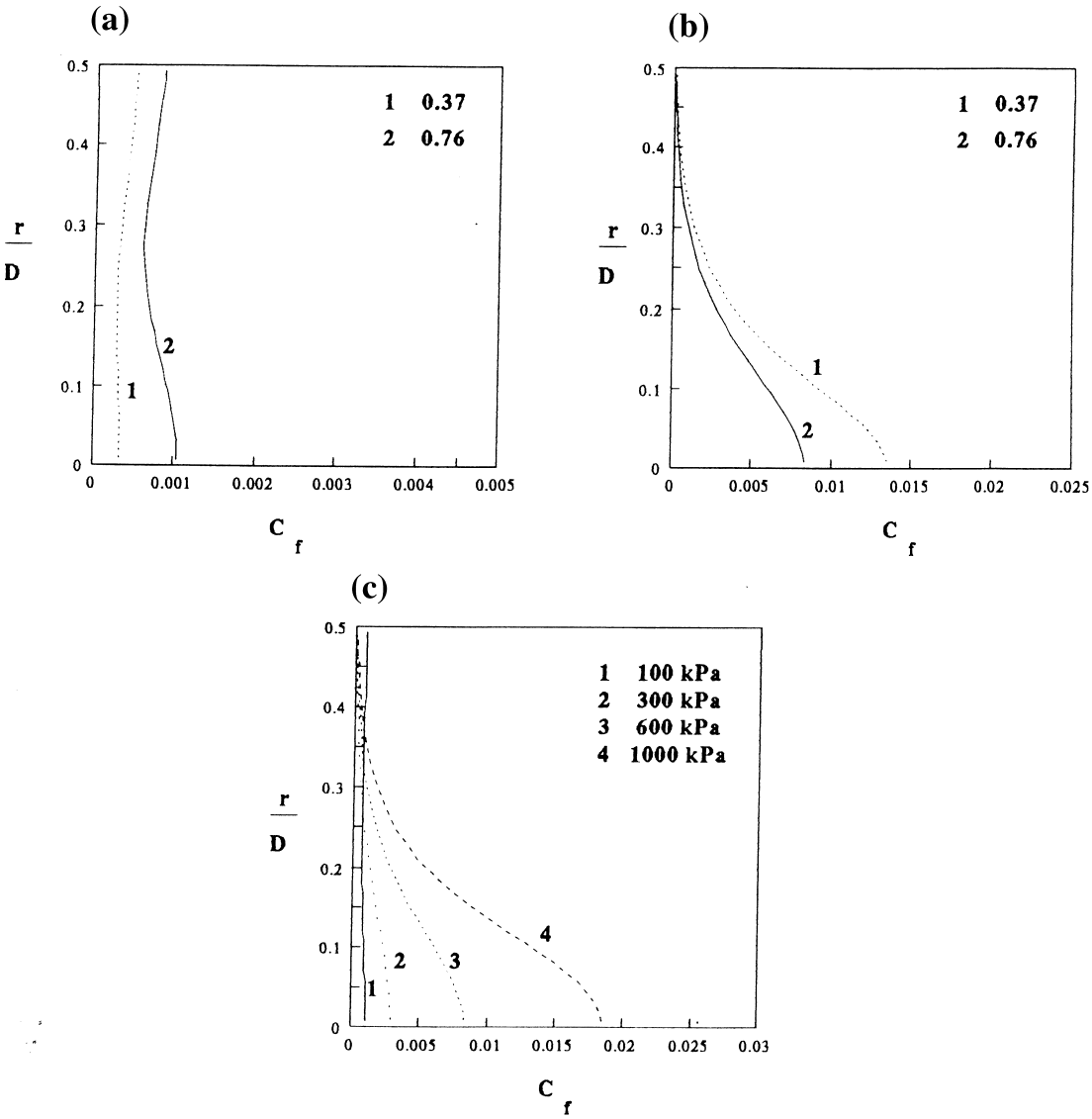

Fig. 7. Exit fuel vapour distribution for different swirl numbers at (a) low combustor pressure (100 kPa), and (b) high combustor pressure (600 kPa). (c) Exit fuel vapour distribution for different combustor pressures (swirl number = 0.76).

4.4. Exit NO concentration

It is observed (Fig. 8a, b) that an decrease in the inlet swirl number increases the bulk NO concentration at the exit which is even more pronounced at lower pressure. However, earlier literature with gaseous fuel reported contrasting results. Although Kevin Owen et al. [6]

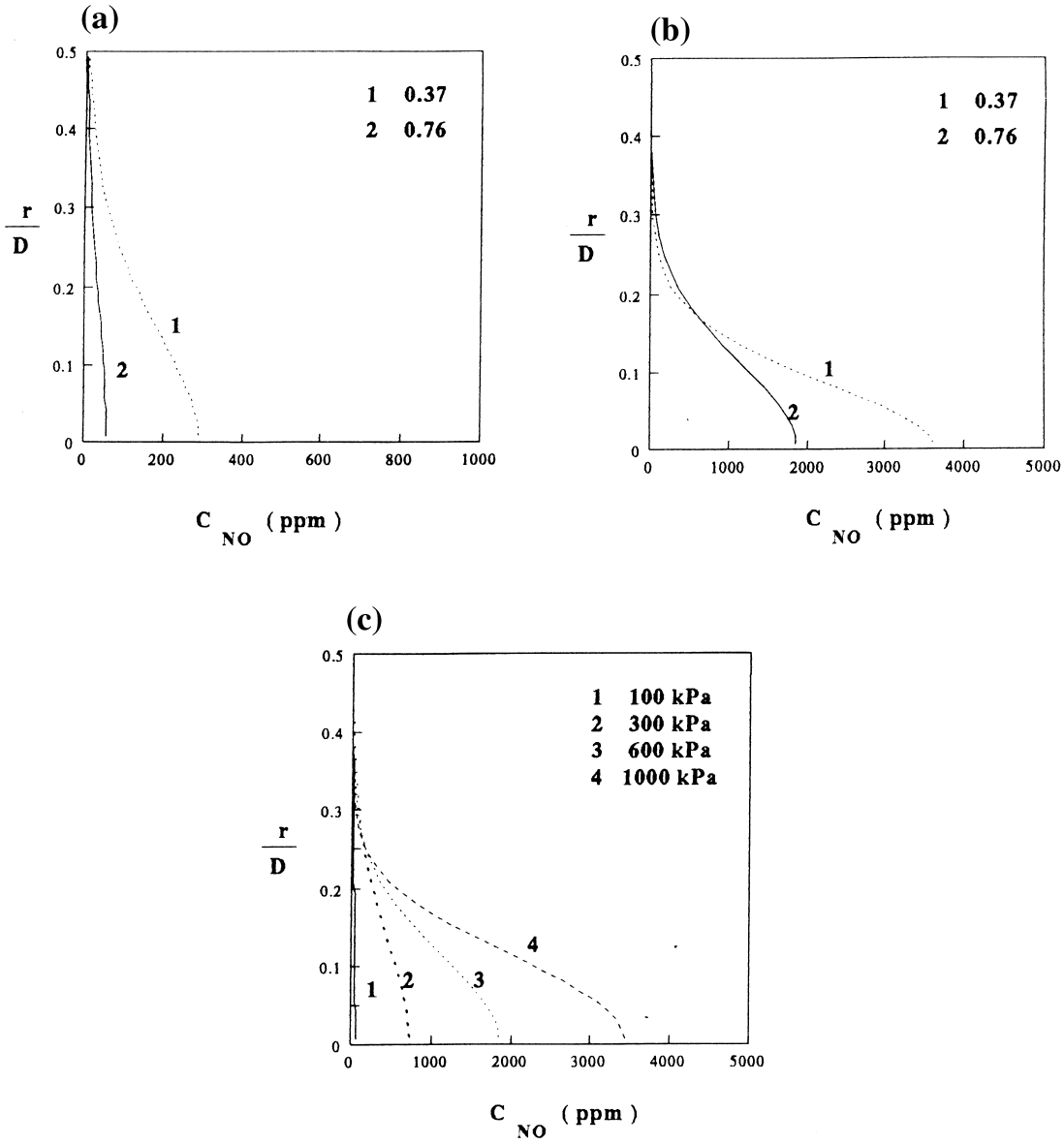

Fig. 8. Exit NO distribution for different swirl numbers at (a) low combustor pressure (100 kPa), and (b) high combustor pressure (600 kPa). (c) Exit NO distribution for different combustor pressures (swirl numbers = 0.76).

reported a significant increase in NO emission with increasing swirl, the works of Ramos [1] and Sokolov et al. [3] showed the opposite trend. The present results are in line with that of Ramos and Sokolov et al., which show that a larger recirculation zone at the combustor centreline causes lower NO emission (Table 1). Since the formation of thermal NO is dependent mostly on the temperature, the NO concentration distribution follows the same pattern as that of temperature.

Combustor pressure is found to have a direct influence on the NO formation (Fig. 8c). Recent work of Amin et al. [5] concludes that for gaseous fuel combustion in an axisymmetric gas turbine combustor NO concentration is proportional to the 1.5 exponent of pressure. Although the combustion mechanism of liquid fuel diffusion flame differs from gaseous fuel diffusion flame, in that the former involves an additional physical process of vapourization of liquid fuel before subsequent mixing and combustion, it is most significant to note from our numerical experiment that even for liquid fuel the exponent of pressure, on which NO concentration depends, varies between 1.2 and 1.7. The reason may be that the kinetics of NO formation is very slow due to high activation energy and is almost uninfluenced by the physical processes before combustion.

The results of the work as discussed in the present section provide the guideline to decide upon the operating parameters of a typical gas turbine combustor to have the desired performances. For an example, a relatively lower inlet pressure but a higher inlet swirl of incoming primary air will result in a better fuel economy, a more uniform exit temperature distribution along with a clean emission. The physical mechanisms through which the operating parameters are related to the performance characteristics depend mainly on the nature of gas phase flow field and its interaction with the dispersed droplet phase as described in the present section. This understanding will help in designing a combustion system to meet the requirements of a desired performance at given operating conditions.

5. Conclusion

A numerical model of liquid fuel spray combustion in a gas turbine combustor has been developed to recognize the influence of inlet swirl number and combustion pressure on important combustion and emission characteristics as follows:

- Influence of inlet swirl number on combustion efficiency shows opposite trends at low and high combustion pressure. An increase in swirl number reduces the combustion efficiency at a combustion pressure of 100 kPa, while it increases the combustion efficiency when the combustor pressure is 600 kPa. However, with an increase in inlet swirl number, pattern factor remains almost unaffected while NO formation is reduced at all pressures.
- An increase in combustor pressure, for a fixed inlet temperature, results in (i) a reduction in combustion efficiency, (ii) further non-uniformity in the exit temperature distribution, and (iii) an increase in the formation of NO in the combustor. The variation of mean NO at exit with pressure is found to have an exponent of around 1.5.

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